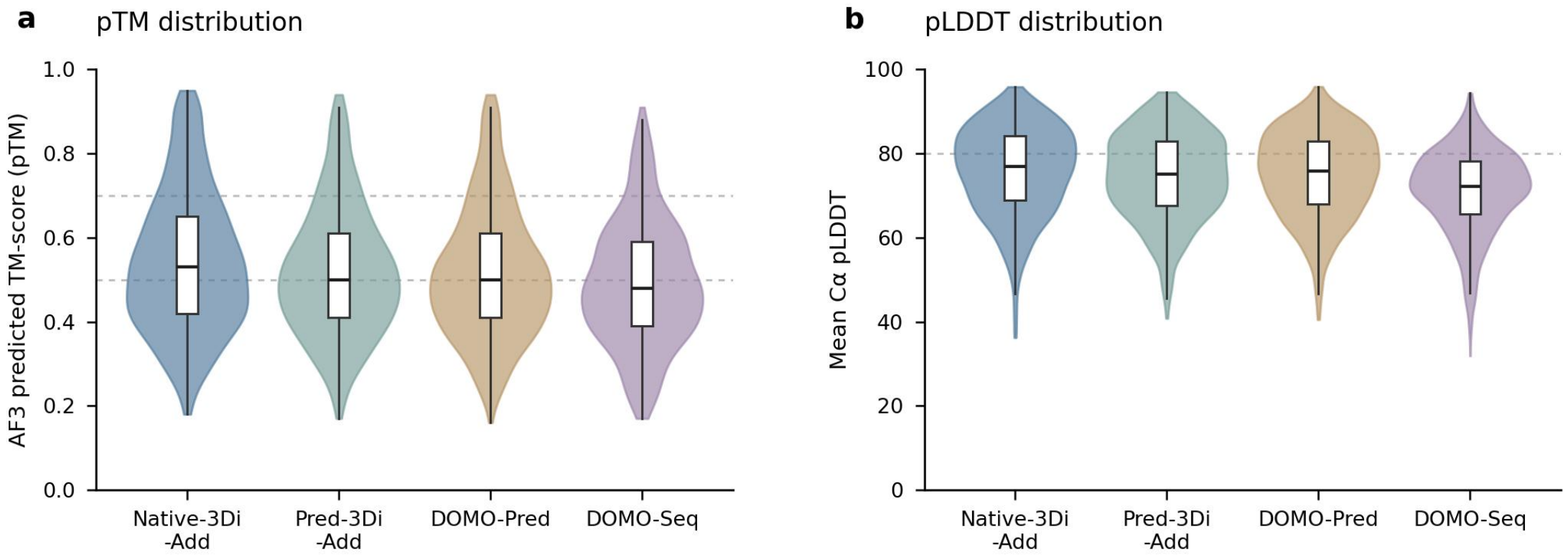
天然序列与多种条件生成序列的结构置信度比较

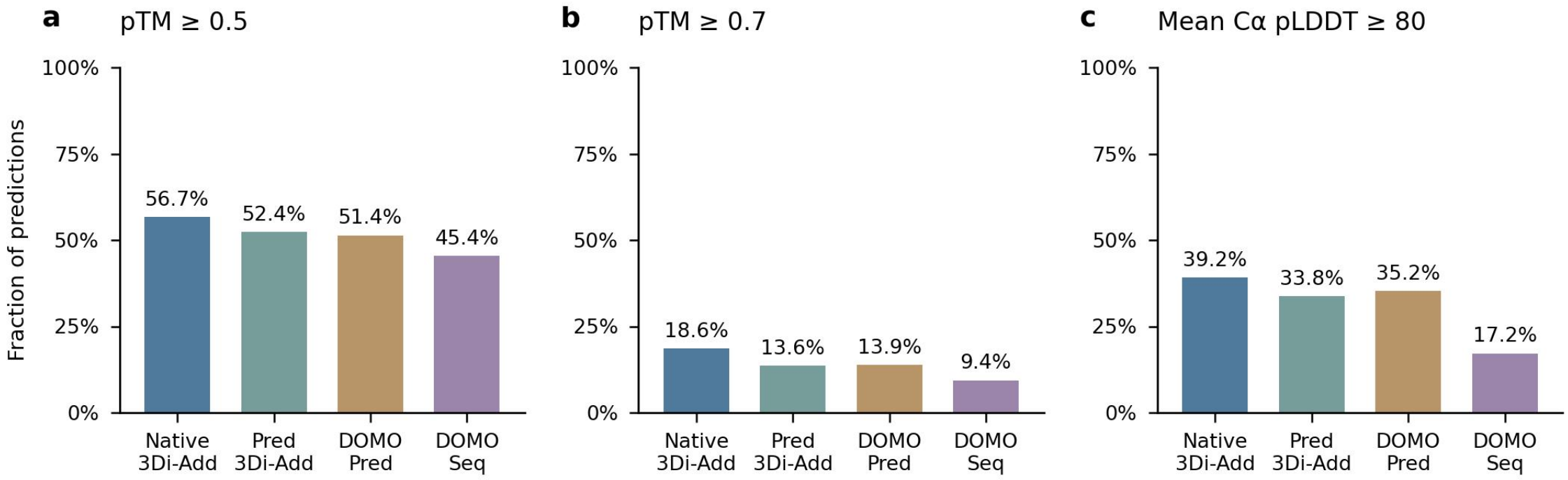
姚昌铭

11月13号

四种方法跑alphafold结果



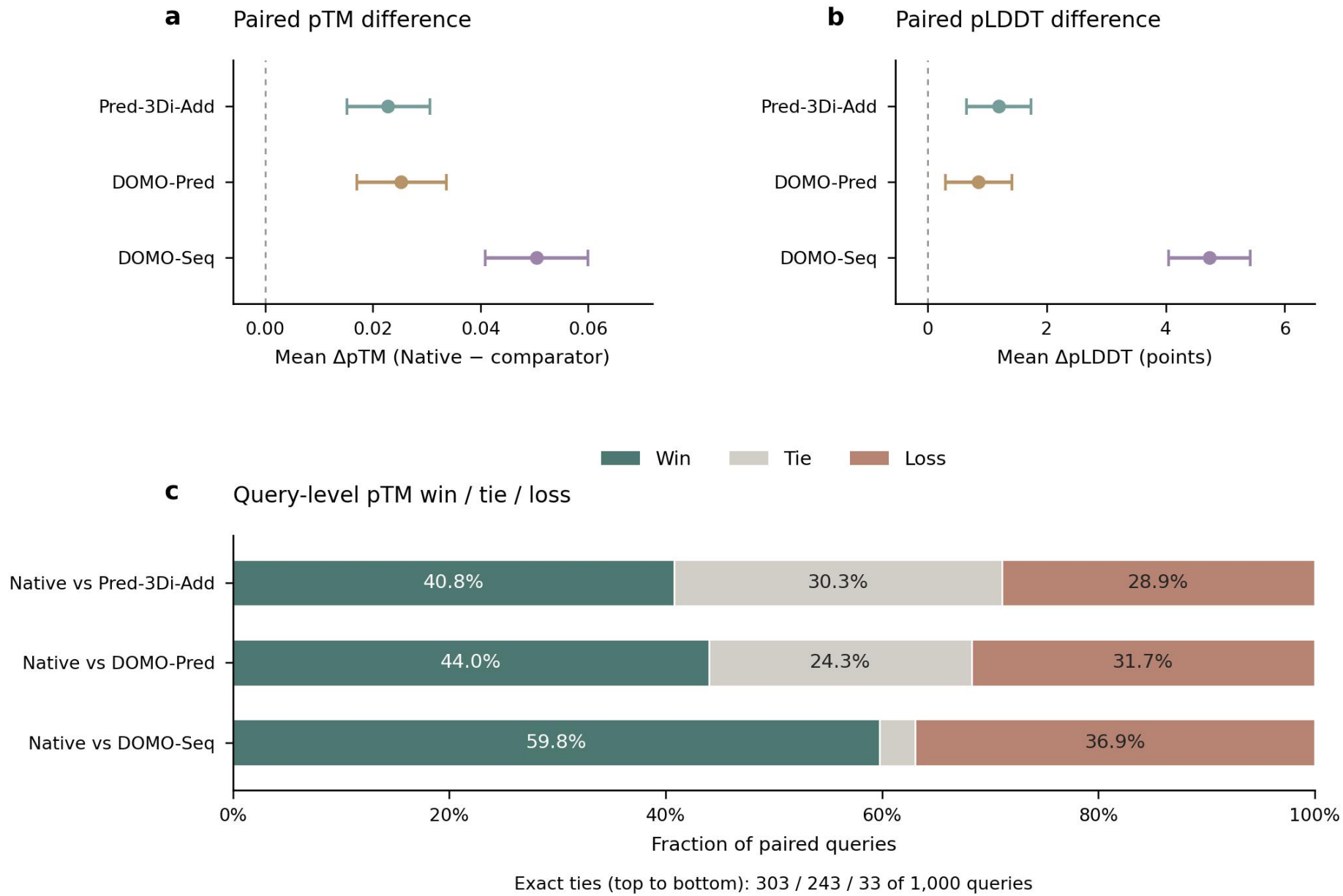
1,000 queries per method · highest-ranking AF3 model per task



Method	N	Mean pTM	Median pTM	Mean pLDDT	Median pLDDT
Native-3Di-Add	1,000	**0.5417**	**0.530**	**75.77**	**76.87**
Pred-3Di-Add	1,000	0.5189	0.500	74.57	75.12
DOMO-Pred	1,000	0.5165	0.500	74.91	75.89
DOMO-Seq	1,000	0.4913	0.480	71.03	72.14

指标下降由DOMIN or DOMO导致?

Condition	Mean pTM	Median pTM	Mean pLDDT	Median pLDDT
Natural	**0.6504**	**0.660**	**80.66**	**84.33**
Oracle DOMO	0.5907	0.590	77.46	77.69
Native-3Di-Add	0.5562	0.545	76.41	77.56
Pred-3Di-Add	0.5228	0.505	74.99	75.71
DOMO-Pred	0.5153	0.505	74.68	75.55
DOMO-Seq	0.5081	0.500	71.86	71.91


a
Paired pTM difference
Pred-3Di-Add
DOMO-Pred
DOMO-Seq
0.00
0.02
0.04
0.06
Mean ΔpTM (Native − comparator)
b
Paired pLDDT difference
Pred-3Di-Add
DOMO-Pred
DOMO-Seq
0
2
4
6
Mean ΔpLDDT (points)
Win
Tie
Loss
c
Query-level pTM win / tie / loss
Native vs Pred-3Di-Add
40.8%
30.3%
28.9%
Native vs DOMO-Pred
44.0%
24.3%
31.7%
Native vs DOMO-Seq
59.8%
36.9%
0%
20%
40%
60%
80%
100%
Fraction of paired queries
Exact ties (top to bottom): 303 / 243 / 33 of 1,000 queries